REAL-WORLD TECLISTAMAB OUTCOMES FROM THE U.S. MIDWEST MYELOMA NETWORK

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INTRODUCTION

Teclistamab is a BCMA-directed bispecific T-cell engager that offers an off-the-shelf option for relapsed/refractory multiple myeloma (RRMM). Real-world outcomes in patients who are comorbid, elderly, and geographically isolated are still being characterized. The Midwest is a high-priority region for study, where a higher proportion of patients live in rural areas and drive long distances for care.

AIM

This multicenter study aims to evaluate the real-world effectiveness and safety of teclistamab in a cohort of RRMM patients across four high-volume academic centers serving large rural referral bases in the Midwest. We also sought to elucidate various dosing strategies being used in clinical practice and how these potentially affect outcomes.

METHODS

This multicenter retrospective study included 159 patients with RRMM treated at Indiana University, University of Wisconsin, Penn State University, and Ohio State University.

- Included patients treated from October 2022 to March 2025
- All patients completed inpatient step-up dosing and transitioned to outpatient maintenance
- Variables collected included:
  - Demographics
  - Clinical characteristics (e.g. myeloma characteristics, treatment history, prior BCMA exposure)
  - Effectiveness data (response rate, time to progression, overall survival)
  - Toxicity

RESULTS

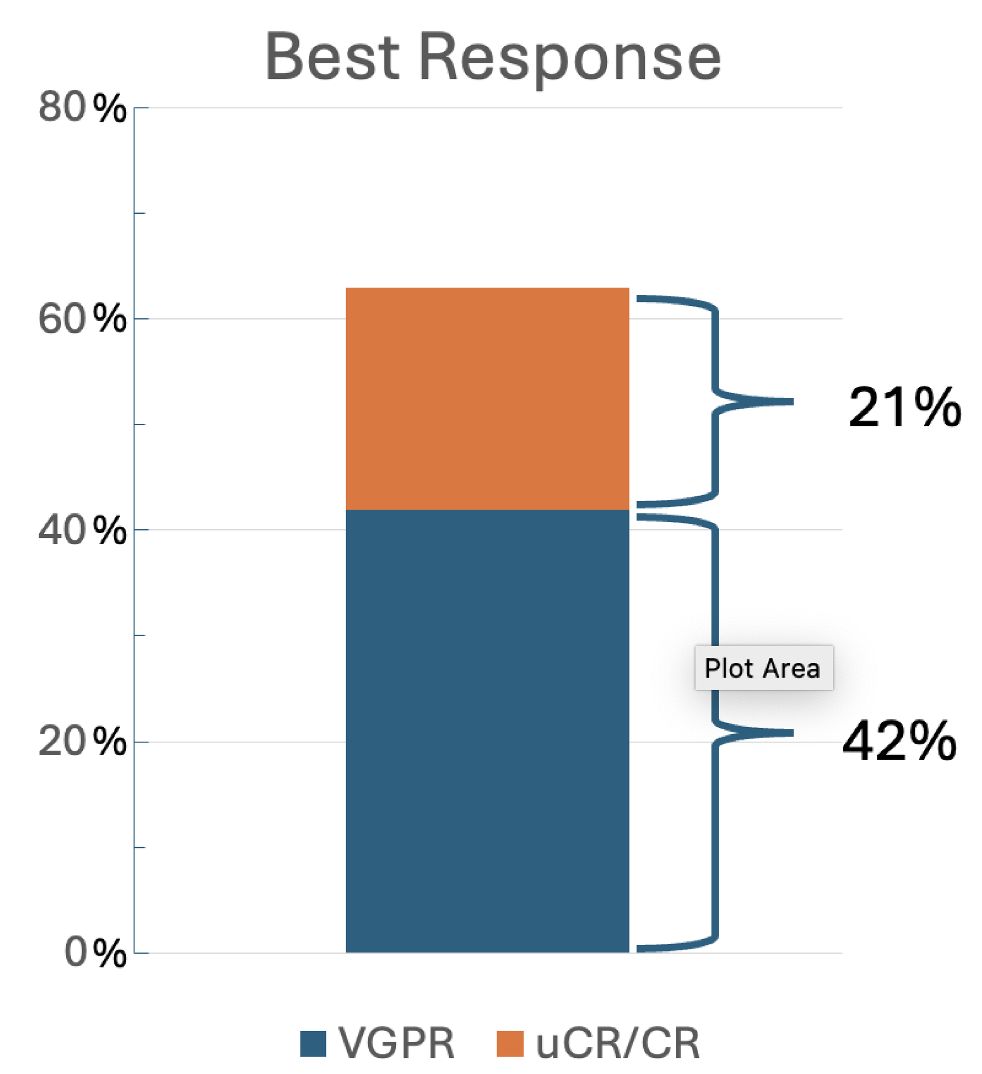

Figure 1. The overall response rate in this cohort was 63%, on par with that seen in MajesTEC-1, with 21% of patients achieving an unconfirmed complete response or complete response (uCR/CR) and 42% of patients achieving a very good partial response (VGPR).

Survival data not shown: Time to progression (TTP) in the overall cohort was 8.2 months. In responders TTP was 15.8 months and in non-responders TTP was 2.1 months. TTP was significantly shorter in patients with prior BiTE/CAR-T exposure vs BCMA naïve, not significantly different with prior ADC exposure.

Table 1. Clinical and demographic characteristics of patients at baseline

Variable	Overall Cohort (N=159)
Median age	74 years
Sex	
Male	53.8%
Female	46.3%
Race	
White	90.7%
Black	6.8%
Asian	1.9%
R-ISS Stage III	29.6%
Myeloma Subtype	
Intact immune globulin	74.5%
Light chain	23.6%
Oligo-/nonsecretory	1.9%
High risk cytogenetics*	29.4%
Extramedullary disease	25.3%
Triple-refractory**	97.5%
Penta-refractory***	51.6%
Prior autologous stem cell transplant	61.6%
Prior exposure to BCMA-directed therapy	30.2%
Renal dysfunction****	64.8%
Hepatic dysfunction****	3.6%
Cardiac dysfunction****	31.6%
Pulmonary dysfunction****	22.8%

Table 2. Adverse events in the cohort

Event	Overall Cohort (N=159)
Any grade CRS	51%
Grade 3 CRS	4%
Any grade ICANS	18%
Grade 3 ICANS	2%
Grade 3 infection	29%
Fatal infection	5%

Table 1 Footnotes

*Includes del(17p), del(13q), t(4;14), t(14;16), t(14;20), amplification 1q

**Defined as refractory to one immunomodulatory agent, one proteasome inhibitor, and one anti-CD38 antibody

***Defined as refractory to both immunomodulatory agents (lenalidomide, pomalidomide), both proteasome inhibitors (bortezomib, carfilzomib), and one anti-CD38 antibody (daratumumab, isatuximab)

****Baseline organ dysfunction defined as creatinine clearance <45 AST or ALT > 2.5x ULN, total bilirubin > 1.5, INR > 1.5; LVEF < 45%; or baseline oxygen dependence

Dose de-escalation patterns:

- 108 patients underwent dose reduction
- 63 patients reduced intensity upon achieving VGPR or better at a median of 10 weeks
  - 92% of these patients remained progression free at 12 months
- 29 patients reduced dosing due to adverse events, primarily infections, at a median of 16 weeks
  - 76% of these patients remained progression free at 12 months

CONCLUSIONS

Teclistamab retains robust efficacy with a manageable safety profile in this real-world cohort of Midwest patients. Response rates approximate those in the pivotal trial that led to drug approval, with responders exhibiting durable benefit. Prior exposure to BCMA targeted T-cell redirecting therapies impacted response to teclistamab, but not exposure to ADC. Dose de-escalation does not appear to compromise real-world effectiveness, which is a critical strategy for rural patients.

REFERENCES

Moreau P, Garfall AL, van de Donk NWCJ, Nahi H, San-Miguel J, Oriol A, et al. Teclistamab in relapsed or refractory multiple myeloma. *N Engl J Med.* 2022;387(6):495-505. doi:10.1056/NEJMoa2203478.

ACKNOWLEDGEMENTS

This work was completed while I was in training at Indiana University. Thank you to my research mentor Attaya Suvannasankha and all of our collaborators across the Midwest. Thank you to our patients who make our work so meaningful.

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